REDD-Only Evidence and NILM Deployment Claims

Stephen Makonin Technical note version 1.1 25 October 2026

REDD helped establish open, reproducible NILM research and remains valuable for benchmark comparisons. **High REDD scores alone do not establish broad deployment performance.** This note addresses the additional evidence needed for claims across households, seasons, and measurement systems.

Limits to address in deployment claims

- **Coverage.** Six homes with short, uneven valid records provide limited evidence across households and seasons. Many correlated samples do not increase household diversity.
- **Timing.** Mains readings are nominally one second apart; circuit readings approximately three seconds. Verify alignment and recording overlap. Upsampling adds no measured information.
- **Missing readings.** Filling, smoothing, excluding, and joining observations can change transitions, labels, and test difficulty, raising or lowering scores.
- **Reference compatibility.** Check active versus apparent power, units, and channel aggregation. A circuit containing several devices is not individual-appliance ground truth.

A case study of this repository's historical importer

The repairs belong to Stephen Makonin's archived 2015 importer, not REDD's acquisition pipeline. The current Python 3 importer does not perform those repairs. The following counts describe the archived log.

Measure and denominator	Calculated result
Exported rows / observed channel-1 mains rows	1,797,457 / 7,056,896 = **25.47%**
Recorded mains substitutions or rebuilds / exported rows	2,858 / 1,797,457 = **0.1590%**
Detailed fallback rebuilds only	**476 events**; median MAIN increase **301**; maximum **5,116** logged power units

No raw-data rerun was performed. Native circuit cadence contributes to selection: 25.47% is neither uptime nor sensor loss. Repair labels are not measured clock errors. Fallback magnitudes cover only those 476 cases. These counts establish no particular accuracy effect.

What credible validation should include

Report channels, units, timestamps, quality masks, preprocessing, valid shared exposure, and activations. Keep test targets out of aggregate repair. Test independently measured mains, held-out homes, and another acquisition system; report appliance energy/event errors and when predictions become available.

Complementary data by purpose. Use **AMPds2** and **R1Hz** for longitudinal energy tests, selected calibrated **UK-DALE** mains for measurement checks, and **REFIT** for household diversity. The choices imply no overall accuracy ranking. Apply the same missingness and quality scrutiny to every dataset; retain R1Hz s/+ flags and score imputed references separately.

Disclosure. Stephen Makonin coauthored the AMPds2, RAE, and R1Hz dataset publications.

Sources and detail REDD original paper (2011); NILMTK diagnostics (2014); DataWrangle_REDD code and log (66ea7e0). Dataset publications: AMPds2 (2016), UK-DALE (2015), REFIT (2017). **R1Hz:** S. Makonin, D. J. Agron, and P. Palmer. Descriptor: The Residential 1Hz Energy Dataset (R1Hz). IEEE Data Descriptions, vol. 3, pp. 1–11, 2026. DOI 10.1109/IEEEDATA.2026.3746420. Full references, calculations, graphs, and limitations appear in the white paper.